

An Analysis of Augmented Reality Aided Vehicle Operation

1st Jaelyn Domingo*

Computer Science Department
California Polytechnic State University
San Luis Obispo, CA, USA
jaelyn.c.domingo@gmail.com

2nd Dimitrije Kostic*

Computer Engineering Department
California Polytechnic State University
San Luis Obispo, CA, USA
dimitrikostic99@gmail.com

3rd Georgia Monahan

Computer Science Department
California Polytechnic State University
San Luis Obispo, CA, USA
georgiamonahan@yahoo.com

4th Krzysztof Pietroszek

Institute of IDEAS
American University Washington DC
Washington DC, USA
pietrosz@american.edu

5th Irene Humer

Computer Science Department
California Polytechnic State University
San Luis Obispo, CA, USA
ihumer@calpoly.edu

6th Christian Eckhardt

Computer Science Department
California Polytechnic State University
San Luis Obispo, CA, USA
eckhardt@calpoly.edu

**These authors contributed equally to this work*

Abstract—In this study, we investigated the effect of augmented reality aided dashboard information projected on the windshield on car driving performance. As augmented reality technology becomes more popular and accessible, essential driving information such as the current velocity can be projected to *where the traffic happens* in terms of location and distance. In comparison with a Heads Up Display (HUD) on a windshield, augmented reality devices offer the possibility to place the data farther in front of the car. Because of this, the eyes of the driver do not need to re-focus constantly to a closer object, when most of the imminent attention needs to be set to the space in front of the vehicle. To test in a safe environment, we developed a virtual reality car parkour and conducted a user study. Splitting our participants in two equally big groups, one with a conservative dashboard and the other with the velocity displayed 10 m ahead and over the horizon. Assessing the participant's performance in areas such as maintaining the speed limit, distance to the traffic, and time to match up with the given speed limit, we found striking positive results using the augmented reality solution.

Index Terms—Augmented Reality, Car, Driving, Immersion

I. INTRODUCTION

Augmented Reality (AR), as an emerging technology, becomes increasingly relevant to enhance our perception of our surroundings with additional data. AR [1] combines the real physical world with computer-generated content. This is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. The visual elements can be placed in any point in space, or they can be entangled with a moving object. Modern UI design allows the visual elements to be non-disruptive (e.g. semi-transparent) in order to not distract from crucial visual feedback such as traffic while driving a car. Conservatively, the operator of a vehicle has all data elements like the speedometer placed on the dashboard. The driver needs to frequently look away from the traffic to the dashboard, re-focus their eyes, process the

information, and change the focal point again back to the traffic [2] [3] [4]. Besides the process time, the readjusting of the eyes requires the convergence adjusting of eye movement [5] [6] [7]. In this work, we investigate how augmented reality can help minimize the time to check the current vehicle parameters in order to enhance safety. Augmented reality devices such as a helmet-mounted device (HMD) or a heads up display (HUD) on the windshield always have a fixed distance and need the eye's focal point to readjust, but the convergence point can be altered for HMD's due to their stereoscopic features [8] [9]. Our approach is to provide the operator with a speedometer which is not only displayed in front rather than below on the display, but also at a greater distance from the car (approx 10 m), where most of the immediate attention is required. By minimizing the speed-checkup time, the operators attention should be elevated. To measure this effect, we developed a Virtual Learning Environment (VLE) in virtual reality (VR) to achieve results under the safety of a simulation. Assessing user data of (complex) physical phenomena, including Newtonian mechanics such as driving a car, with the help of a VLE utilizing VR proved to be successful for Gravitational Waves [10], Special Relativity [11] and other topics [12] [13] [14]. As an emerging technology, VLE's [15] [16] offer a unique possibility to convey information in an immersive, interactive way, including depth perception in a 3D environment [17] [18]. VLE's were thoroughly studied for their effectiveness in different fields such as Computer Science [19], Chemistry [20] and many more [21] [22] [23]. Furthermore, we conducted a user study including the the Game Experience Questionnaire (GEQ) [24] as well as the Game Immersion Questionnaire (GIQ) [25] to assess the level of immersion of our participants and in order to cross reference the results.

II. METHODOLOGY

For this study, we created an immersive driving environment which prompted participants to drive around a VR landscape. We asked participants to follow a guided path and respect speed and ethical driving rules. The goal was to assess their speed maintenance and collision avoidance for two groups: One with an augmented reality speedometer and one with standard dashboard.

With a volunteer participant size of $n = 39$, we conducted a 3-phase study. First, a pre-questionnaire was used to assess demographics and prior VR experience. Secondly, participants entered the VR simulation. We measured participants' deviation from the speed limit, close following distances, and collisions with traffic. Lastly, each participant answered a post-questionnaire to determine their level of immersion with the GEQ and GIQ.

A. Car Driving Simulation

1) *Development*: Our driving simulation immerses participants by entering them into a VLE in which they are seated in a car. We designed our VLE using Unity with OpenVR and SteamVR Unity package, supporting our participants HMD's: Oculus Rift, HTC Vive Pro, and Valve Index. The participants controlled their vehicle using a keyboard for gas, brakes, and steering wheel. To enhance the immersion, a steering wheel animation was implemented in the direction of the turn. Two modes were utilized using one of two speedometer types: an augmented speedometer aligned with traffic, out in front of the moving vehicle (see Fig. 1) and a standard speedometer on the dashboard. These modes were assigned randomly to participants.



Fig. 1. Augmented Reality speedometer in city traffic: In the center the augmented speedometer in orange lettering is clearly visible. Car following the Blue Overhead Guided Path while maintaining 70 km/h speed limit regulated in the city.

Unity's free asset store items were used to create a highway, a small city, speed limit signs, and non-player cars. The highway was used to get accustomed using the controls to drive. The small city is populated with many buildings and has increased traffic. The non-player cars move at a constant speed along a static path to simulate traffic.

Speed limit signs are placed along the guided path. Along the highway, the 100 and 70 km/h zones are utilized. In the city, there 70 and 50 km/h zones as well as a school crossing zone, which enforces a 30 km/h speed limit, and the end

zone, which suggests the player has completed the course, can stop the vehicle, and exit the simulation. Transitions to slower speeds advised to the driver allowed them to take the curves in the road with more control.

Lastly, a static path was used as a guide to direct the driver throughout the VLE as is shown by the blue line above the street in Fig. 1.

2) *Simulation Priorities*: We made two priorities for the participants to follow while driving. These priorities were essential to keep participants on course and driving ethically in order to get consistent driving data. The first priority was to maintain the driving speed, which was used to make drivers aware of the speedometer and the regulated speed limit and require them to utilize that information while driving. The second priority was to follow the blue overhead path, which was used to keep drivers on course and get them to the end zone (see Fig. 1). The driver's use of the speedometer was essential to get conclusive results on how the speedometer's location affected the participant's awareness about their driving speed.

B. Data Collection

During the simulation, we collected participants' driving data. This included tracking their current speed, speed limit zone, following distance to other cars, and collisions. This data was stored in a CSV file for us to analyze.

C. Questionnaires

Each participant was required to complete a questionnaire prior and post their driving experience. Their level of immersion was documented and driving metrics stored. For the immersion assessment, we used the GEQ and GIQ.

1) *Pre-questionnaire*: This questionnaire was directed to gain insight of the participants' demographics and prior VR experience.

2) *Post-questionnaire*: Participants responded to the GEQ and the GIQ, which asked questions relating to immersion both during and after our simulation in this questionnaire. The immersion questions prompted participants to rank their experience on a scale of 0 to 4, with the descriptions being (0) "Not at all", (1) "Slightly", (2) "Moderately", (3) "Fairly", and (4) "Extremely". The GEQ asked questions about their experience "during" the VLE to rank their immersion in regards to competence, sensory and imaginative immersion, flow, tension or annoyance, challenge, and positive or negative effects. The GIQ asked questions about their experience "after" the VLE about attention, temporal dissociation, transportation, emotional involvement, challenge, and enjoyment.

III. DEMOGRAPHICS

The participants demographic information was collected by the pre-questionnaire, which included participants' employment status, level of education, ethnicity, household income, age, and gender. Our test group consisted of $n = 39$ participants with 48% of participants in the augmented group and 52% participants in the standard group. The statistics of the participant's demographics are shown in Fig. 2.

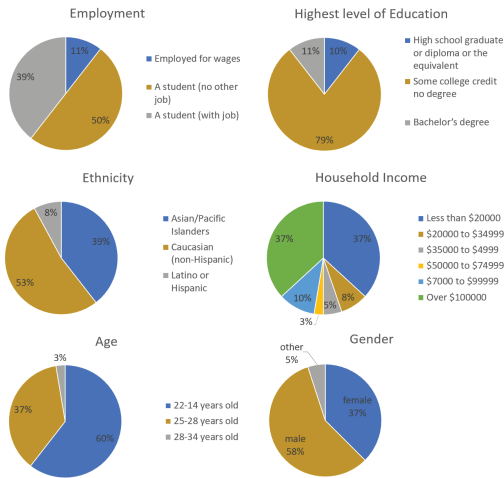


Fig. 2. Group demographics. The distribution of the participants level of education, age, employment status, ethnicity, gender and annual household income.

IV. RESULTS

Throughout the driving simulation we recorded the driver's average speed, current speed zone, close following distances, and collisions. Using the driver's average speed per second and the posted speed limit, we found the driver's deviation from the speed limit. We observed this result by comparing the driver's deviation from the posted speed limit and total number of collisions throughout their drive. Furthermore, we assessed the immersion of the participants with the Game Experience Questionnaire and the Game Immersion Questionnaire. We, then, used this immersion feedback along with our driver data to find correlations between the two data sets.

A. Total Driving Data

The following graphs present the average driving data from all participants, concentrating on the two driver control categories: speed-deviation and collisions, and average speed throughout the simulation in specific speed zones. The graphs compare our two driver groups' (augmented vs. standard) performance in these data categories.

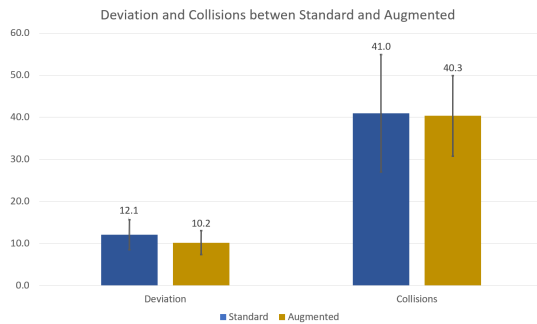


Fig. 3. Average speed-deviation (km/h) from posted speed limit and number of collisions for all participants: (blue) using standard speedometer and (gold) using augmented speedometer

According to Fig. 3, the average driver's deviation from the speed limit, measured in km/h, while using the augmented speedometer was slightly, but notably less than using the standard speedometer. We found an average of speed-deviation of 12.05 for the standard group with a standard deviation of 3.55 and 10.16 for the augmented group with a standard deviation of 2.84. This data shows a better speed handling for the augmented group over the standard group. On average, participant's who completed the simulation with the standard dashboard deviated 18.6% further from the posted speed limit.

Additionally, Fig. 3 shows that the total amount of collisions made by the average driver was remarkably similar between the two groups. Furthermore, the data indicated on average, drivers in the augmented group made 41.0 collisions, while the drivers in the standard group made 40.3 collisions while proceeding through the course. With standard deviations exceeding at least 9 on average, the type of speedometer did not affect how the drivers maneuvered the car.

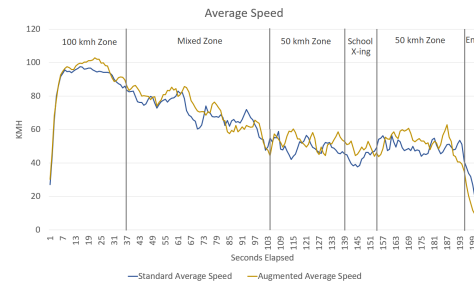


Fig. 4. Average speed curves for all participants throughout the simulation: (blue) using the standard speedometer and (gold) using the augmented speedometer. Zones included 100 km/h, mixed zone, 50 km/h, End Zone, and a school crossing. The mixed zone includes several rapid changes between 100,70 and 50 km/h. Vertical lines signify the moment the car changed speed zones.

Between participants using the standard vs. augmented speedometers, the overlay of average speeds in Fig. 4 shows the trend of the Augmented (Gold) group to be higher, and closer to the posted speed limits than the Standard (Blue) group. The mixed zone includes several rapid speed changes. However, we assessed the speed difference for every participant in every frame accurately. In conjunction with the previous data indicated in Fig. 3, the drivers in the Augmented group trended higher, driving faster and closer to the posted speed limit. The faster and more accurate speed also caused the speed at then end to trend to 0 km/h several seconds faster in augmented-dash vehicles, as they completed the guided path in a shorter time.

As starkly illustrated in Fig. 5, participants using the augmented display were able to drastically cut down the time it took to transition from their previous speed to the new regulated speed. For example, moving from 100 to 50 km/h took 5.8 seconds less, from 50 to 100 km/h in 3.7 seconds less, and from 50 to 70 and 70 back to 50 km/h took 4 and 2.6 seconds less, respectively. A change in speed zones in our VLE usually signified a change in the road—whether it be a driver merging from highway to road, or making a turn. During these

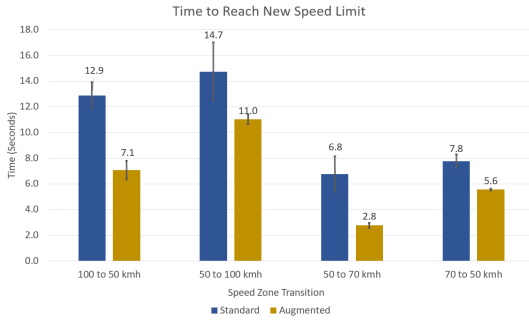


Fig. 5. Time taken (in seconds) to reach within 5 km/h of the current speed limit when changing from one Speed Zone to another with (Blue) using the standard speedometer and (Gold) using the augmented speedometer

situations it is clear the augmented display greatly helped in drivers' awareness and maintenance of the correct speed.

B. Immersion Responses

Our post-questionnaire asked participants to evaluate the degree of immersion experienced in the VLE during and after the driving simulation. The questionnaires assessed a different immersion category to calculate the simulation's ability to engage the participant.

The GIQ designated immersion responses which participants reported experiencing while completing the driving simulation. The GEQ, which collected responses subsequent to engagement, used six categories of immersion: basic attention, temporal dissociation, transportation, challenge post, emotional involvement, and enjoyment.

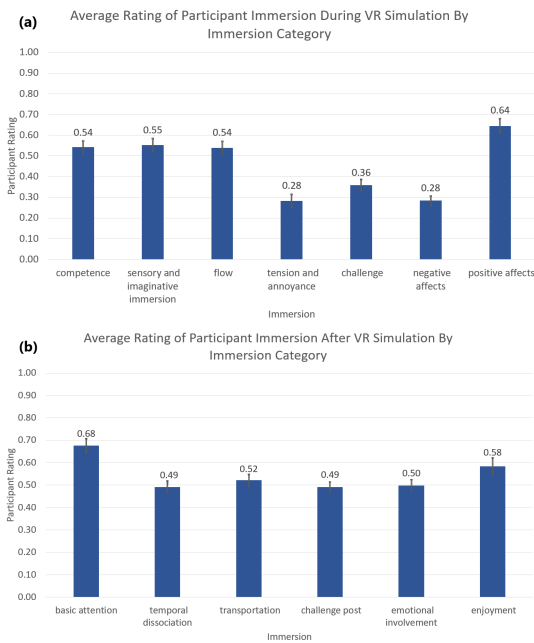


Fig. 6. Average Immersion feedback from all participants during and after the simulation

Fig. 6 (a) shows the participant's assessment of their experience while engaged in the Virtual Learning Environment. Based on the results from the participants, we assigned ratings for each category within the range 0 (not at all) to 1 (extremely). Overall, the results reflected that the average participant was fully immersed in the experience, as indicated by high marks particularly in flow and sensory and imaginative immersion. Although the data shows most participants felt competent in controlling the car and that the simulation fostered a positive environment, it is also evident that a good amount of participants felt challenged and indicated slight frustration.

The average participant's evaluation of immersion after completing the simulation is illustrated by Fig. 6 (b). We observed that the average participant was focused while driving through the course and enjoyed their experience. Furthermore, the results show that the average participant felt immersed, indicating that they were transported into the VLE and were motivated to complete the course to the best of their ability.

C. Immersion Correlations

To compare our results from the immersion feedback to our driver data, the engagement results were correlated with the ability to control the car of the drivers who performed in the top third of the group, from both participant groups.

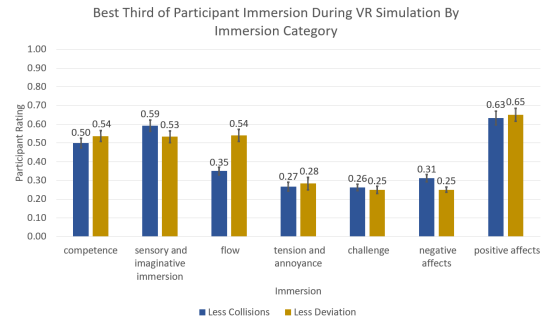


Fig. 7. Immersion feedback from participants during the simulation for the best third of drivers with less collisions and less speed-deviations

Specific subsets of user data was chosen in order to find correlations between performance and immersion. According to Fig. 7, the top third best drivers felt less frustration, challenge and negative emotions throughout their drive in the VLE. A trend between the top drivers in the two car control categories can be seen: the immersion categories competence, sensory and imaginative immersion, and positive affects were important in promoting ethical car control. We also noticed that flow was an important factor, particularly, for drivers that had a lower average speed-deviation. Overall, the best drivers experienced these immersion factors more.

The immersion feedback from participants after the simulation can be seen in Fig. 8. Participants in the best driver group reported higher basic attention and enjoyment, which appears to have aided in completing the course, maintaining the proper speed and following distance to traffic. Transportation,

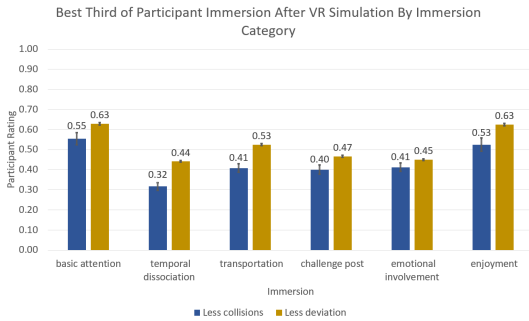


Fig. 8. Immersion feedback from participants after the simulation for the best third of drivers

although comparable to other categories, proved to be another significant component for fostering an immersive environment for participants, specifically, in the group with less speed-deviation. Therefore, attention, enjoyment, and transportation, were categories of immersion that provided us with our driving results for drivers in the top third based on performance.

V. CONCLUSION AND FUTURE WORK

We developed a VLE for testing augmented reality features in a traffic environment for car drivers. In a user study, we conducted experiments to hold a certain speed and avoid traffic obstacles for two subgroups: one group with augmented speedometer in front of the participants and the other group with a standard dashboard. Our results show an increased awareness of the traffic surroundings and a greater control over speed for augmented-reality-aided participants in our user study. AR-supported participants further showed a quicker transition to the allowed speed. Future work includes the use of an AR device such as the Hololens to conduct a user study under real world conditions.

REFERENCES

- [1] K. Rathinavel, H. Wang, A. Blate, and H. Fuchs, "An extended depth-at-field volumetric near-eye augmented reality display," *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 11, pp. 2857–2866, 2018.
- [2] M. Sodhi, B. Reimer, and I. Llamazares, "Glance analysis of driver eye movements to evaluate distraction," *Behavior research methods, instruments, computers : a journal of the Psychonomic Society, Inc.*, vol. 34, pp. 529–38, 12 2002.
- [3] D. Crundall and G. Underwood, "Chapter 11 - visual attention while driving: Measures of eye movements used in driving research," in *Handbook of Traffic Psychology* (B. E. Porter, ed.), pp. 137–148, San Diego: Academic Press, 2011.
- [4] T. Langner, D. Seifert, B. Fischer, D. Goehring, T. Ganjineh, and R. Rojas, "Traffic awareness driver assistance based on stereovision, eye-tracking, and head-up display," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3167–3173, 2016.
- [5] G. Underwood, P. Chapman, N. Brocklehurst, J. Underwood, and D. Crundall, "Visual attention while driving: sequences of eye fixations made by experienced and novice drivers," *Ergonomics*, vol. 46, no. 6, pp. 629–646, 2003. PMID: 12745692.
- [6] G. Prabhakar and P. Biswas, "Improving eye gaze controlled car dashboard using simulated annealing," 01 2017.
- [7] R. Jose, G. A. Lee, and M. Billingham, "A comparative study of simulated augmented reality displays for vehicle navigation," in *Proceedings of the 28th Australian Conference on Computer-Human Interaction, OzCHI '16*, (New York, NY, USA), p. 40–48, Association for Computing Machinery, 2016.
- [8] Wai-Tat Fu, J. Gasper, and S. Kim, "Effects of an in-car augmented reality system on improving safety of younger and older drivers," in *2013 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 59–66, 2013.
- [9] M. Tonniss, C. Sandor, G. Klinker, C. Lange, and H. Bubb, "Experimental evaluation of an augmented reality visualization for directing a car driver's attention," in *Fourth IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR'05)*, pp. 56–59, 2005.
- [10] S. Lontschar, K. Pietroszek, I. Humer, and C. Eckhardt, "An immersive and interactive visualization of gravitational waves," in *2020 6th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 178–184, 2020.
- [11] G. Chu, I. Humer, and C. Eckhardt, "Special relativity in immersive learning," in *Immersive Learning Research Network* (D. Beck, A. Peña-Rios, T. Ogle, D. Economou, M. Mentzelopoulos, L. Morgado, C. Eckhardt, J. Pirker, R. Koitz-Hristov, J. Richter, C. Gütl, and M. Gardner, eds.), (Cham), pp. 16–29, Springer International Publishing, 2019.
- [12] T. Brown, J. Lomsdalen, I. Humer, and C. Eckhardt, "Immersive learning for scale and order of magnitude in newtonian mechanics," in *Immersive Learning Research Network* (D. Beck, A. Peña-Rios, T. Ogle, D. Economou, M. Mentzelopoulos, L. Morgado, C. Eckhardt, J. Pirker, R. Koitz-Hristov, J. Richter, C. Gütl, and M. Gardner, eds.), (Cham), pp. 30–42, Springer International Publishing, 2019.
- [13] S. K. Babu, S. Krishna, U. R., and R. R. Bhavani, "Virtual reality learning environments for vocational education: A comparison study with conventional instructional media on knowledge retention," in *2018 IEEE 18th International Conference on Advanced Learning Technologies (ICALT)*, pp. 385–389, 2018.
- [14] J. Jimenez, D. Gutierrez, and P. Latorre, "Gaze-based interaction for virtual environments," *Journal of Universal Computer Science*, vol. 14, pp. 3085–3098, nov 2008.
- [15] Z. Pan, A. Cheok, H. Yang, J. Zhu, and J. Shi, "Virtual reality and mixed reality for virtual learning environments," *Computers & Graphics*, vol. 30, pp. 20–28, 02 2006.
- [16] V. Callaghan, M. Gardner, B. Horan, J. Scott, L. Shen, and M. Wang, "A mixed reality teaching and learning environment," pp. 54–65, 08 2008.
- [17] A. Abdoli Sejzi, "Augmented reality and virtual learning environment," *Journal of Applied Science Research (JASR)*, vol. 11, pp. 1–5, 05 2015.
- [18] G. Burdea and P. Coiffet, *Virtual Reality Technology*. Wiley - IEEE, Wiley, 2003.
- [19] C. K. Mosquera, A. Steinmaurer, C. Eckhardt, and C. Guetl, "Immersively learning object oriented programming concepts with school," in *2020 6th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 124–131, 2020.
- [20] D. Fujiwara, K. Kellar, I. Humer, K. Pietroszek, and C. Eckhardt, "Vsepr theory, an interactive and immersive virtual reality," in *2020 6th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 140–146, 2020.
- [21] D. Beck, "Special issue: Augmented and virtual reality in education: Immersive learning research," *Journal of Educational Computing Research*, vol. 57, no. 7, pp. 1619–1625, 2019.
- [22] S. Hewawalpita, S. Herath, I. Perera, and D. Meedeniya, "Effective learning content offering in moocs with virtual reality - an exploratory study on learner experience," *Journal of Universal Computer Science*, vol. 24, pp. 129–148, feb 2018.
- [23] S. Ternier, R. Klemke, M. Kalz, P. van Ulzen, and M. Specht, "Arlearn: Augmented reality meets augmented virtuality," *Journal of Universal Computer Science*, vol. 18, pp. 2143–2164, aug 2012.
- [24] W. IJsselstein, Y. de Kort, and K. Poels, *The Game Experience Questionnaire*. Technische Universiteit Eindhoven, 2013.
- [25] C. Jennett, A. L. Cox, P. Cairns, S. Dhoparee, A. Epps, T. Tijs, and A. Walton, "Measuring and defining the experience of immersion in games," *International Journal of Human-Computer Studies*, vol. 66, no. 9, pp. 641 – 661, 2008.